Alcohol, Phenol & Ether-1

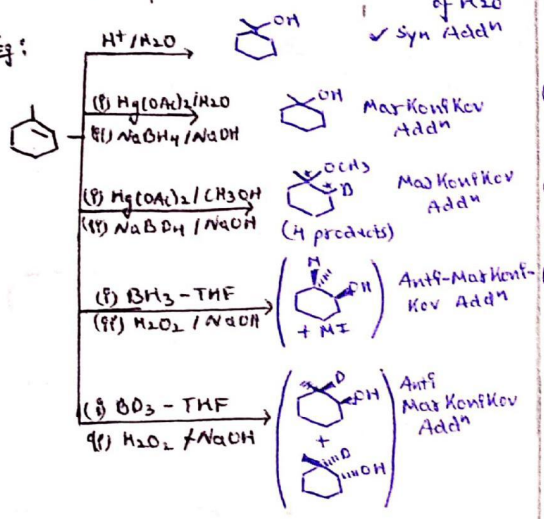
Glycerol) Important Trihydric Alcohol.

#Method of Preparation of Alcohol:

1) Hydrocarbons to Alcohol ⇒

Dil. H_2SO_4 (Hydration of Alkene)	Oxymercuration Demercuration (ONDM)	Hydroboration Oxidation (HBO)
✓ H^+/H_2O	✓ (i) $Hg(OAc)_2/H_2O$ (ii) $NaBH_4/NaOH$	✓ (i) BH_3-THF (ii) $H_2O_2/NaOH$
✓ Carbocation intermediate	✓ Non-classical carbocation	✓ Non-classical carbocation
✓ Rearrangement	✓ No Rearrangement	✓ No Rearrangement
	✓ Mar Konikov addn of H_2O	✓ Anti-Mar Konikov addn of H_2O
		✓ Syn Addn

Eg:

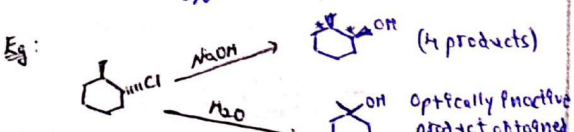


2) Alkyl Halide to Alcohol ⇒

$$R-X \xrightarrow[S_N2]{NaOH} R-OH \quad (1^\circ, 2^\circ)$$

$$R-X \xrightarrow[S_N1]{H_2O} R-OH \quad (2^\circ, 3^\circ)$$

Eg:



3) From Reduction of Carbonyl Compounds ⇒

→ Reduction will be done by metal Hydrides.

$M^+ H^\ominus$ → $Nu^\ominus$ → Every $Nu^\ominus$ has tendency to react in the priority order as

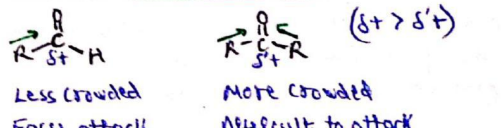
(i) Acid-Base Rxn
(ii) Rxn with π bond
(iii) Rxn with σ-bond

Some σ-bonds behave as π, for ex. bonds of cyclopropane △

→ Ease of Attack of $H^\ominus$ depends on

- ↳ Electrophilicity of compound (More δ+)
- ↳ Steric Hinderence (Should be less crowded)

Therefore, attack of $H^\ominus$ on aldehyde > Ketone

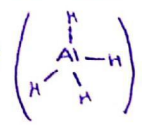


$$R-\overset{O}{\overset{\|}{C}}-H \xrightarrow[(ii)\ H_2O \rightleftharpoons NH_4Cl]{(i)\ H^\ominus} R-CH_2-OH \quad (1^\circ\ Alc.)$$

$$R-\overset{O}{\overset{\|}{C}}-R \xrightarrow[(ii)\ H_2O \rightleftharpoons NH_4Cl]{(i)\ H^\ominus} R-\overset{OH}{\overset{|}{C}}H-R \quad (2^\circ\ Alc)$$

Reducing Agents ⇒

1) $LiAlH_4$ (LAH)

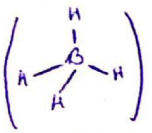


(+) More Reactive due to weak Al-H bond

(+) Never taken with PPS like H_2O due to high reactivity.

(+) Reduces all except
- (i) Alcohol
- (ii) Amine
- (iii) Alkene

2) $NaBH_4$ (SBH)

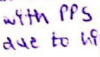


(+) Comparitively less Reactive due to strong B-H bond

(+) Can be taken with any kind of Solvent

(+) Can Reduce Only
- (i) Aldehyde
- (ii) Ketone
- (iii) Acid Halide
- (iv) Imine (-N=C)

3) DIBAL-H



(+) Only 1 $H^\ominus$ donor

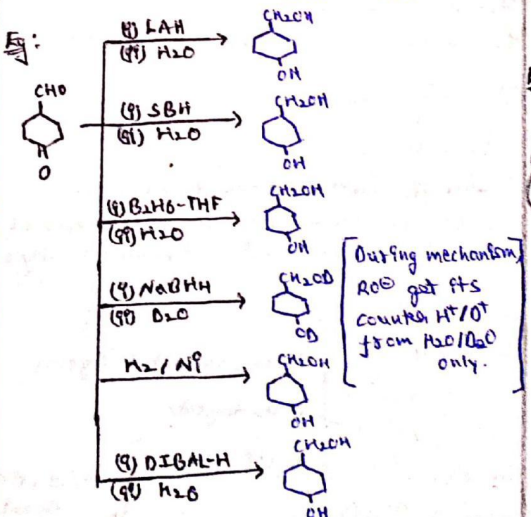
4) B_2H_6 - THF

(+) Syn addn of H

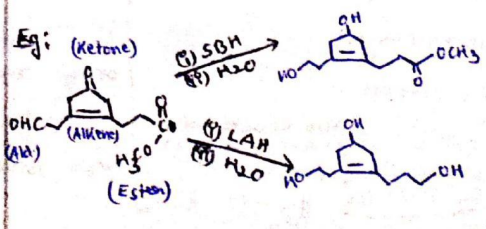
5) H_2/Catalyst

H_2/Pd H_2/Pt H_2/Ni $H_2/Ra\text{-}Ni$

Eg:



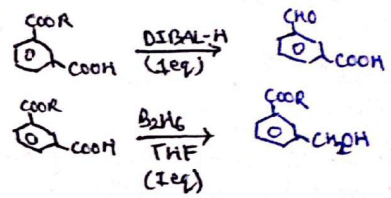
Eg: (Ketone)



For DIBAL-H, R·O·R: $R-\overset{O}{\overset{\|}{C}}-OR' > R-\overset{O}{\overset{\|}{C}}-OH$

For B_2H_6, R·O·R: $R-\overset{O}{\overset{\|}{C}}-OR' < R-\overset{O}{\overset{\|}{C}}-OH$

Eg:



If above Reagents are taken in excess, then both -COOR & -COOH would get reduced into $-CH_2OH$ & -CHO

4) From Reduction of Carboxylic Acid & Esters ⇒

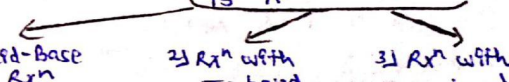
$$R-COOR' \xrightarrow[\text{or } H_2/Catalyst]{LAH \text{ or } B_2H_6-THF} R-CH_2OH + R'-OH$$

$$R-COOH \xrightarrow{\quad\quad} R-CH_2OH + H_2O$$

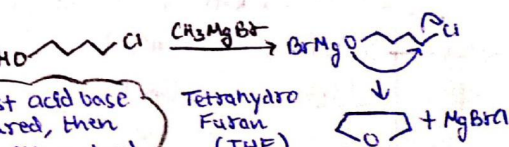
5) From Grignard Reagent :→

$$R-X + Mg \longrightarrow \overset{\delta-}{R}-\overset{2s^+}{Mg}-\overset{\delta-}{X}$$

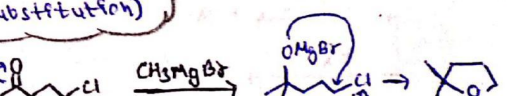
Here there is no R^-, there is R^δ



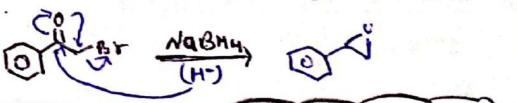
Eg:



Eg:



Eg:



NOTE: No. of moles of $RMgX$ = (No. of lg +1) consumed

1) $H-\overset{O}{\overset{\|}{C}}-H$ (Formaldehyde) $\xrightarrow[(ii)\ H_2O]{(i)\ RMgX}$ $H-\overset{OH}{\overset{|}{C}}H-R$ (1° ROH) (1 mol RMgX)

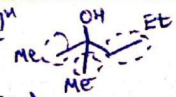
2) $R-\overset{O}{\overset{\|}{C}}-H$ (Aldehyde) → $R-\overset{OH}{\overset{|}{C}}H-R$ (2° ROH) (1 mol RMgX)

3) $R-\overset{O}{\overset{\|}{C}}-R$ (Ketone) → $R-\overset{OH}{\overset{|}{C}}-R$ (3° ROH) (1 mol RMgX)

4) $R-\overset{O}{\overset{\|}{C}}-lg$ → $R-\overset{OH}{\overset{|}{C}}-R$ (2 mol RMgX)

Q. Carbonyl compound + $R'MgBr$ → 

Find possible pairs of G.R. and Carbonyl comp.

Sol^{n} 

Total diff Groups = 2 (Me & Et)

(i)  + CH_3MgBr

(ii)  + $EtMgBr$

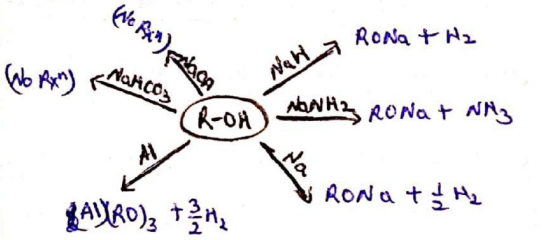
Ans

Chemical Properties of Alcohol :

1) Acidic Nature of Alcohol :→

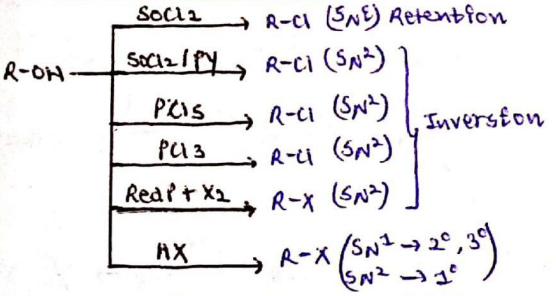
→ Reaction will always proceed from strong acid to weak acid.

→ Thus R-OH will show acidic nature only if acidic specie formed after rx^{n} is weaker.



(i) Acidity of H_2O and H_2CO_3 is more than R-OH, therefore R-OH does not react with NaOH & $NaHCO_3$.

2) Substitution Reaction :→



3) Lucas Test (Dry HCl / Anhyd. AlX_3) :→

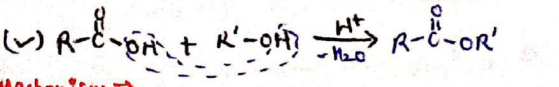
3° R-OH $\xrightarrow[\text{Anhyd. } AlX_3]{HCl}$ R-Cl (Imediate white Turbidity)

2° R-OH $\longrightarrow$ R-Cl (White Turbidity in 5-7 min)

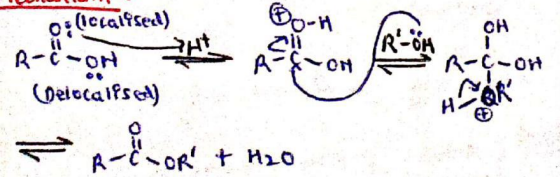
1° R-OH $\longrightarrow$ R-Cl (No Turbidity at Room Temperature)

4) Esterification Reaction :→

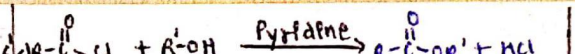
It is formation of fruity smelling esters when Alcohol reacts with acid and it derivative (acid halid & acid anhydride).

(v) 

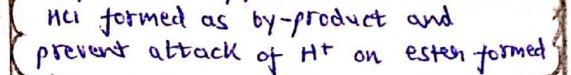
Mechanism ⇒



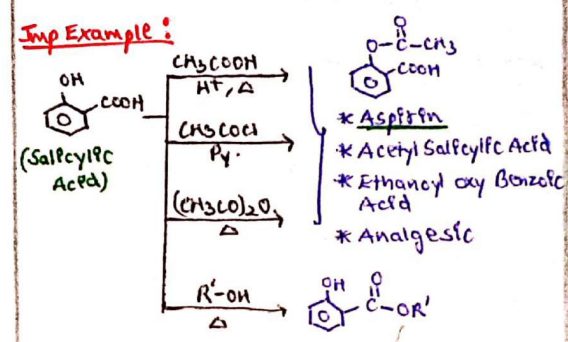


(vi) 

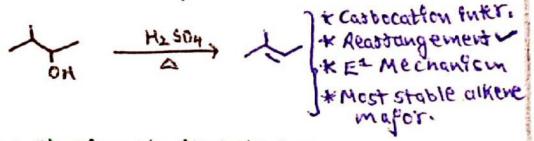
Heavy base like pyridine used to consume HCl formed as by-product and prevent attack of H^+ on ester formed

(vii) 

Imp Example :

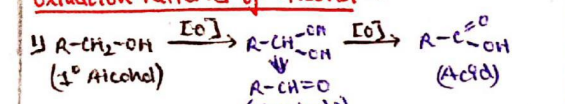


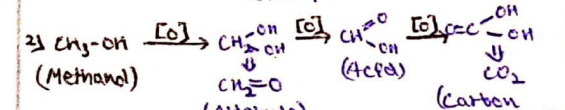
5) Dehydration of Alcohol :→

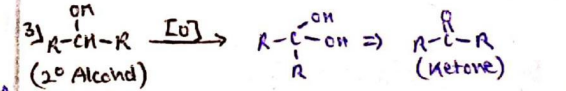


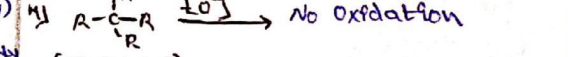
6) Oxidation of Alcohols :→

Oxidation Patterns of Alcohol ⇒

1) 
(1° Alcohol) (Aldehyde) (Acid)

2) 
(Methanol) (Aldehyde) (Acid) (Carbon dioxide)

3) 
(2° Alcohol) (Ketone)

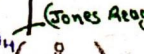
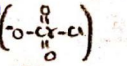
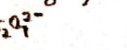
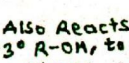
4) 
(3° Alcohol)

-OH jis carbon atom se attached hai, uske har ek -H ko -OH se substitute kar do in successive oxidative steps.

Oxidising Agents ⇒

1) $KMnO_4$, Δ
2) $K_2Cr_2O_7$, Δ
3) CrO_3 / H_2SO_4 (Jones Reagent)

→ Strong Oxidising Agents

4) PCC 
5) PDC 
6) Collins Reagent 
7) X_2 / NaOH
8) NBS → Also Reacts with 3° R-OH, to form alkene
9) Cu, 300°C / Δ 

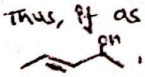
→ Mild oxidising Agents

Cr (VI)	→	Cr (III)
orange		green
Mn (VII)	→	Mn (IV)
Purple		Brown

Order of pH of OA ⇒

Jone Reagent < PCC < PDC < Collin's Reagent

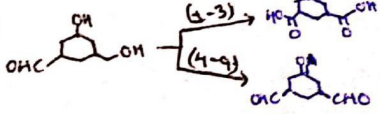
Jone Reagent → Acidic; PDC → Almost Neutral; Collin's Reagent → Basic

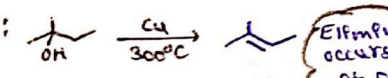
Ques Thus, if asked best OA for oxidation of 

Ans We choose either PDC or Collin's Reagent because acidic OA can disturb alkene.

Strong OA
- 1° ROH → Acid
- CH_3OH → CO_2
- 2° ROH → Ketone
- 3° ROH → No Oxd^{n}

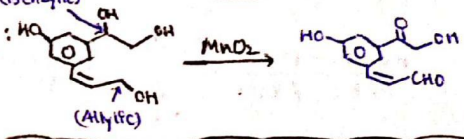
Mild OA
- 1° ROH → Aldehyde
- CH_3OH → Formaldehyde
- 2° ROH → Ketone
- 3° ROH → No Oxd^{n}

Eg: 

Eg: 
Elimination occurs instead of oxidation.

Special oxidising Agent (MnO_2) ⇒

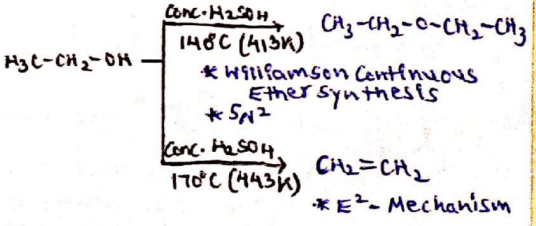
MnO_2 → Oxidises only allylic and Benzylic -OH

Eg: 

NOTE: MnO_2 is also a mild oxidising agent

Method of Preparation of Ether :

1) Dehydration of Alcohol :→

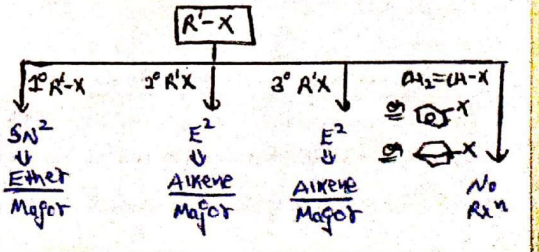


→ 2° & 3° alcohols are not suitable for synthesis of ether by this process.

2) Williamson Ether Synthesis :→

Sodium Alkoxide and Alkyl Halid react to form ether in this rx^{n}, by S_N2 Mech^{n}.

$RO^-Na^+ + R'-X \longrightarrow R-O-R'$



Short Trick ⇒ जो 1° होगा, उसमें -X दिखा होगा।

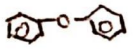
Eg: Predict suitable pair of alkoxide and alkyl halide for following ethers.

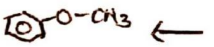
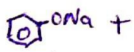
1) $H_3C-CH_2-O-C(CH_3)_2-CH_3$ ← $\wedge X$ + $\wedge ONa$

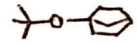
2) $H_3C-O-CH(CH_3)_2$ ← CH_3-X + $\wedge ONa$

3) 1° 1° ← $\wedge X$ + $\wedge ONa$
Less Crowded, More Crowded
Decided by rate of S_N2 at both possible 1° carbons

4) $\wedge O-\wedge$ ← No such pair possible

5)  ← No such pair possible

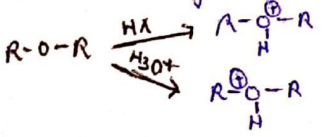
6)  ←  + CH_3-X

7)  ← No such pair possible

#Chemical Properties of Ether:

Cleavage of Ether by HX | Hydrolysis of Ether (H^+/H_2O)

→ In both reaction, attack of proton (H^+) is the first step. But after, rxn would proceed according to following cases.



Case-01 :→

Stable(+) $\wedge O(H)-R$ → X^- → R-OH + (Stable)-X ; → H_2O → R-OH + (Stable)-OH

↓ 3° alkyl ⇒ Resonance

Case-02 :→

$(1^\circ R)-O(H)-CH_3$ → X^- → CH_3-X + $(1^\circ R)-OH$; → H_2O → CH_3-OH + $(1^\circ R)-OH$

S_N2 (less preffered), S_N2 (more preffered)

Case-03 :→

$(1^\circ R)-O(H)-(2^\circ R)$ → X^- → $(1^\circ R)-X$ + $(2^\circ R)-OH$; → H_2O → $(1^\circ R)-OH$ + $(2^\circ R)-OH$

S_N2

NOTE: (i) If not specified, then by default assume cold HI & form (RI + ROH)

(ii) If Hot HI mentioned, then no alcohol formed. Both RI & R'-I are obtained.

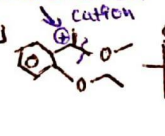
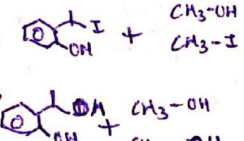
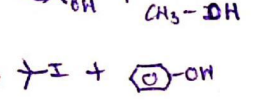
Alcohol Phenol & Ether-2

1) 3° 2° → conc HI → $\wedge I$ + $\wedge OH$; → H_3O^+ → $\wedge OH$ + $\wedge OH$

2) 1° 1° → conc HI → $\wedge I$ + $\wedge OH$; → H_3O^+ → $\wedge OH$ + $\wedge OH$

3) $\wedge O-CH_3$ → conc HI → CH_3-I + $\wedge OH$; → H_3O^+ → CH_3-OH + $\wedge OH$

4) $\wedge O-\wedge$ → conc HI → No Rxn ; → H_3O^+ → No Rxn

5) Reso. Stabilised cation:  → conc HI →  + CH_3-OH + CH_3-I ; → H_3O^+ →  + CH_3-OH + CH_3-OH

6) $\wedge O-\wedge$ → conc HI → $\wedge I$ + $\wedge OH$; → H_3O^+ → $\wedge OH$ + $\wedge OH$

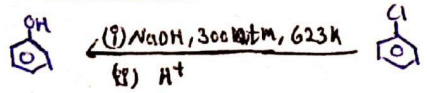
#Physical Properties:

Boiling Point ∝ No. of -OH Groups ; ∝ Molecular Mass ; ∝ $\frac{1}{\text{Branching}}$

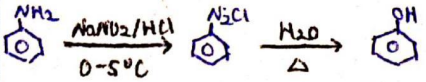
Solubility in H_2O ∝ No. of -OH Groups ; ∝ $\frac{1}{\text{Molecular Mass}}$; ∝ Branching

#Method of Preparation of Phenol:

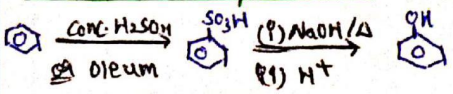
1) Dow's Process :→


(i) NaOH, 300 atm, 623 K ; (ii) H^+

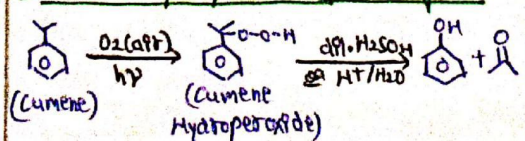
2) From Benzene Diazonium Salt :→


$NaNO_2/HCl$, 0-5°C ; H_2O, Δ

3) From Benzene Sulphonic Acid :→

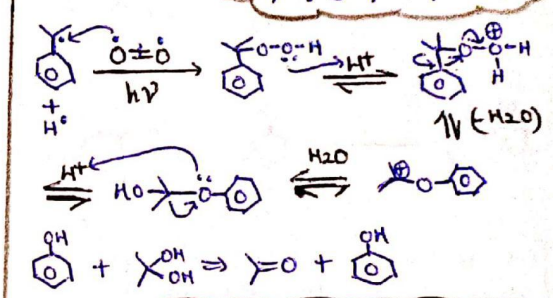

Conc. H_2SO_4 (Oleum) ; (i) NaOH/Δ ; (ii) H^+

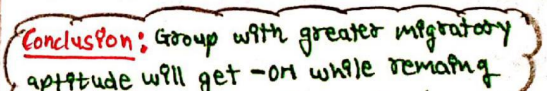
4) Cumene Hydroperoxide Rearrangement :→


(Cumene) ; O_2(air), hν ; (Cumene Hydroperoxide) ; dil. H_2SO_4, H^+/H_2O

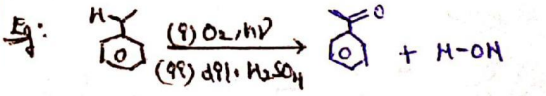
Mechanism :

Migratory Aptitude
H > 3° > 2° > -Ph > 1°


hν ; H^+ ; (-H_2O) ; H_2O

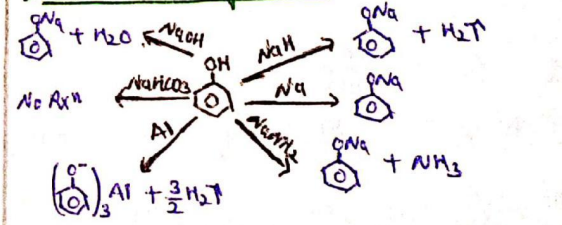


Conclusion: Group with greater migratory aptitude will get -OH while remaing will become carbonyl compound

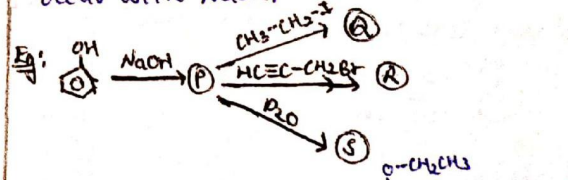
Eg: 
(i) O_2, hν ; (ii) dil. H_2SO_4 ; + H-OH

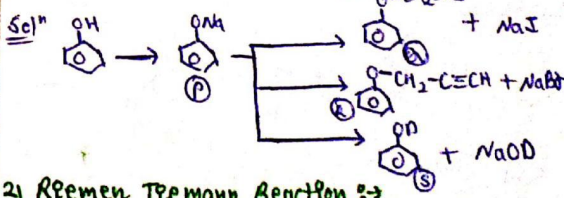
#Chemical Properties of Phenol:

1) Acidic Nature of Phenol :→

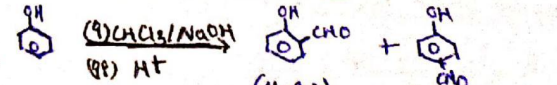

NaOH ; NaH ; Na ; $NaHCO_3$; No Rxn ; Al ; $NaNH_2$; + H_2↑ ; + NH_3

→ Phenol is more acidic than water but less acidic than carbonic water (H_2CO_3), thus no rxn with $NaHCO_3$ while rxn occur with NaOH.

Eg: 
NaOH ; CH_3-CH_2-I ; $HC\equiv C-CH_2Br$; D_2O

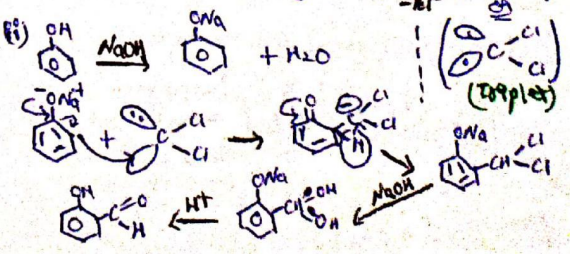
Sol^n 
+ NaI ; + $NaBr$; + NaOD

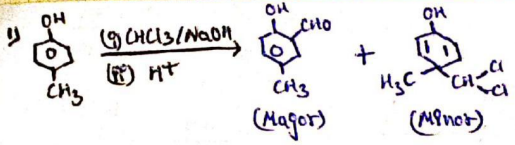
2) Reemen Teemann Reaction :→

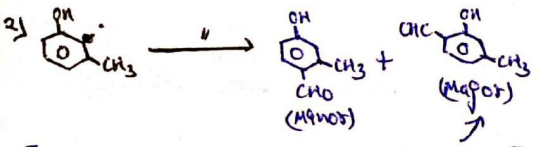

(i) $CHCl_3$/NaOH ; (ii) H^+ ; (Major) ; (Minor)

Mechanism →

(i) $CHCl_3 + NaOH \rightarrow Na^+\bar{C}Cl_3 + H_2O$; (rds) ; $-\bar{C}l$; (Singlet)

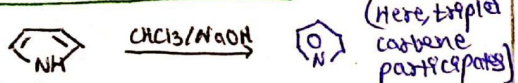
(ii) 
NaOH ; $+H_2O$; (Triplet) ; H^+ ; NaOH

1) 

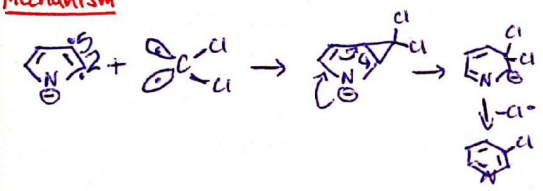
2) 

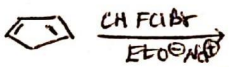
[Due to steric hinderance, carbene attacked at left ortho of the ring]

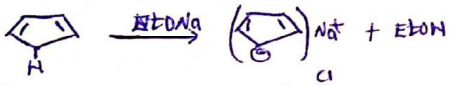
Abnormal Reimar Tiemann ⇒

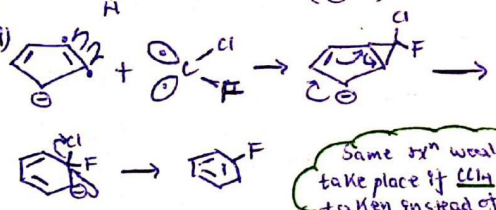


Mechanism



Eg: 

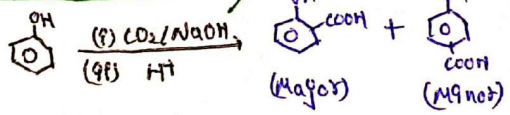
Soln (i) 

(ii) 

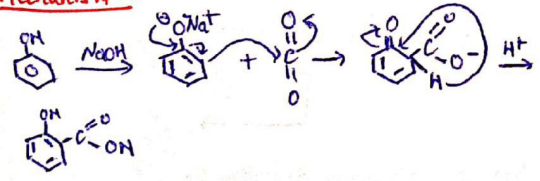


Same rxn would take place if CCl_4 taken instead of CO_2

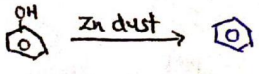
3) **Kolbe Reaction :→**

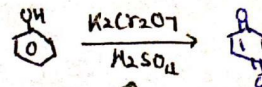


Mechanism ⇒

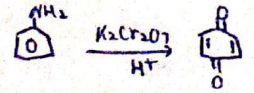


4) **Some other Reactions :→**

(i) 

(ii) 

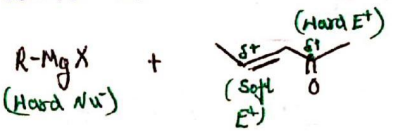
Same rxn is shown by Aniline as well.



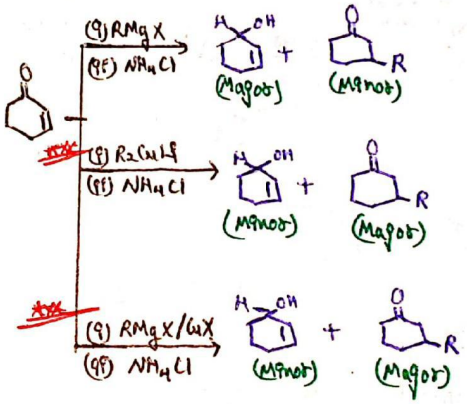
Means, ring should be activated for this rxn to take place.

Exception in Behaviour of Grignard Reagent :

→ GR is a Hard nucleophile and in presence of (C=C) and (C=O), it prefer reacting with (C=O) in normal conditions.



→ But in presence of 'Copper salts (R_2CuLi/CuX)', grignard reagent behaves as a soft nucleophile.



In last two cases, reagent reacted with (C=C) instead of (C=O) and behaved as soft Nu^-.